Finding primary research

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Finding primary research before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Reading a research claim

Without opening the old lesson, explain one key idea from Reading a research claim. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Finding primary research trace and record the first step you can explain without help.

Use W2: Trace the worked case for Finding primary research. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

Individual reasoning check

> Check the proposed Finding primary research explanation against the supplied facts before discussing it.

Use the worked case for Finding primary research. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

A search result describes a benchmark score, but the linked paper uses a different dataset. What should the comparison record contain?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Finding primary research facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

W2 Trace the worked case

> Trace the worked case. Apply the supplied Finding primary research facts and keep your reasoning.

A blog says experiment had100 cases; original methods say40 and supplementary version says60 after revision. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

W3 Audit the limit

> Audit the limit. Apply the supplied Finding primary research facts and keep your reasoning.

A report omits this qualification: Being primary does not guarantee soundness; methods and **evidence** still require assessment. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Finding primary research facts and keep your reasoning.

A search result describes a benchmark score, but the linked paper uses a different dataset. What should the comparison record contain?

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Finding primary research facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Finding primary research to Reproducible AI Research Study, identify a limit and choose the next useful test.

Add a labelled finding primary research **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of finding primary research.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Finding primary research

> Find the source behind a summary. A blog describes a fictional benchmark improvement and links an original study report. Explain the labelled step before checking the changed condition.

Find the source behind a summary. A blog describes a fictional benchmark improvement and links an original study report.

Blog says "90% reliable". Original report records 9/10 correct on one fixed fictional task set and describes its labels.

AI ACADEMY / AI-7/3

Finding primary research

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Follow the primary link

Open the original report rather than treating the blog as the measurement.

↓

2. Locate methods and results

Identify the ten cases, labels and nine correct results.

↓

3. Record source details

Keep report version and page or table reference with the exact scoped result.

CHANGE ONE THING

The original report is unavailable and only the blog remains.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Separate what the blog claims from what the original report records. Which methods and denominator must be inspected, and what status is honest if the original report is unavailable?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Reproducible AI Research Study record.

Project: Reproducible AI Research Study

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Find the source behind a summary. A blog describes a fictional benchmark improvement and links an original study report.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Keep report version and page or table reference with the exact scoped result.

1. Record source details

2. Locate methods and results

3. Follow the primary link

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Record source details Keep report version and page or table reference with the exact scoped result.

Primary research exposes the procedure and **observation**s behind a summary. It still needs scrutiny; being original does not make every interpretation correct.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The original report is unavailable and only the blog remains. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. Mark the measurement as unverified rather than filling in absent methods.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Mark the measurement as unverified rather than filling in absent methods. The summary cannot supply details that have not been inspected.

Repair a missing building block

Review Reading a research claim (ai-7/2). Goal: Explain the **mechanism** and terminology of reading a research claim. Short practice: Calculate the baseline and new rates for 2/10 and 4/10.

Explain both twice the rate and the percentage-point change, and say why neither measures every learner’s improvement.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

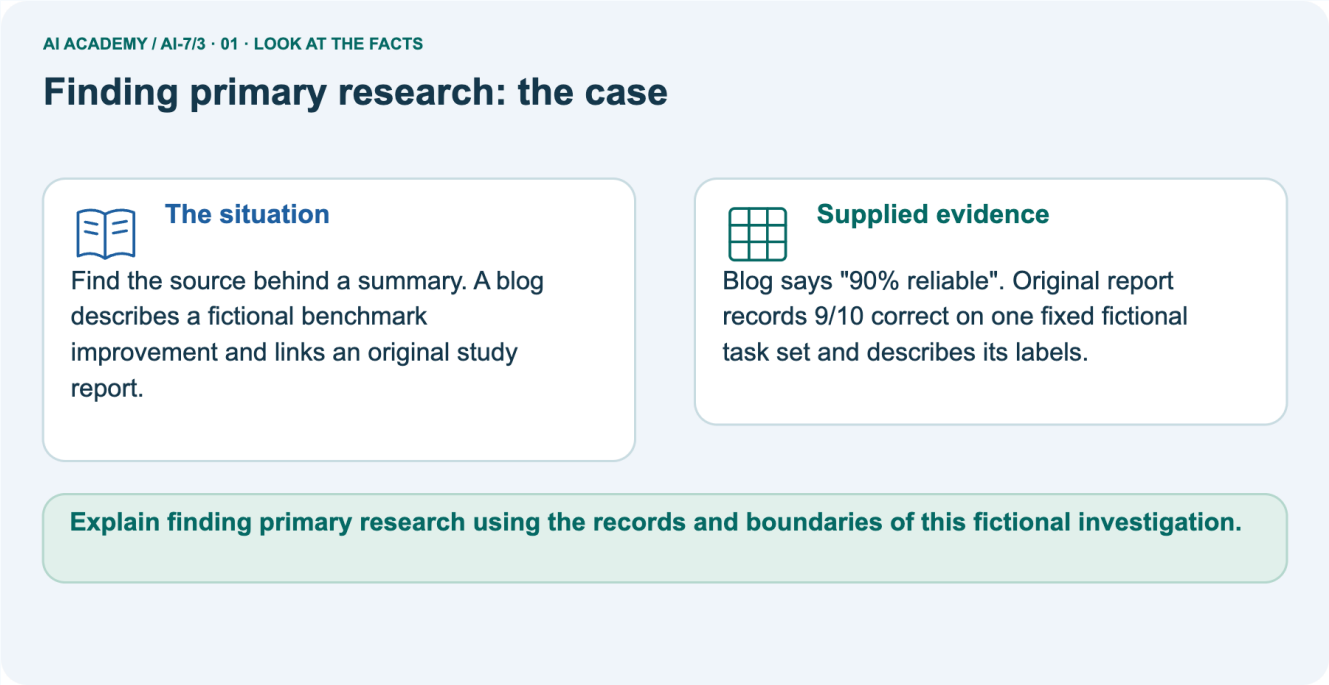
What I repaired, the evidence I kept and the next test

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Finding primary research picture from its labels, then record a relevant limit.

01. Finding primary research: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

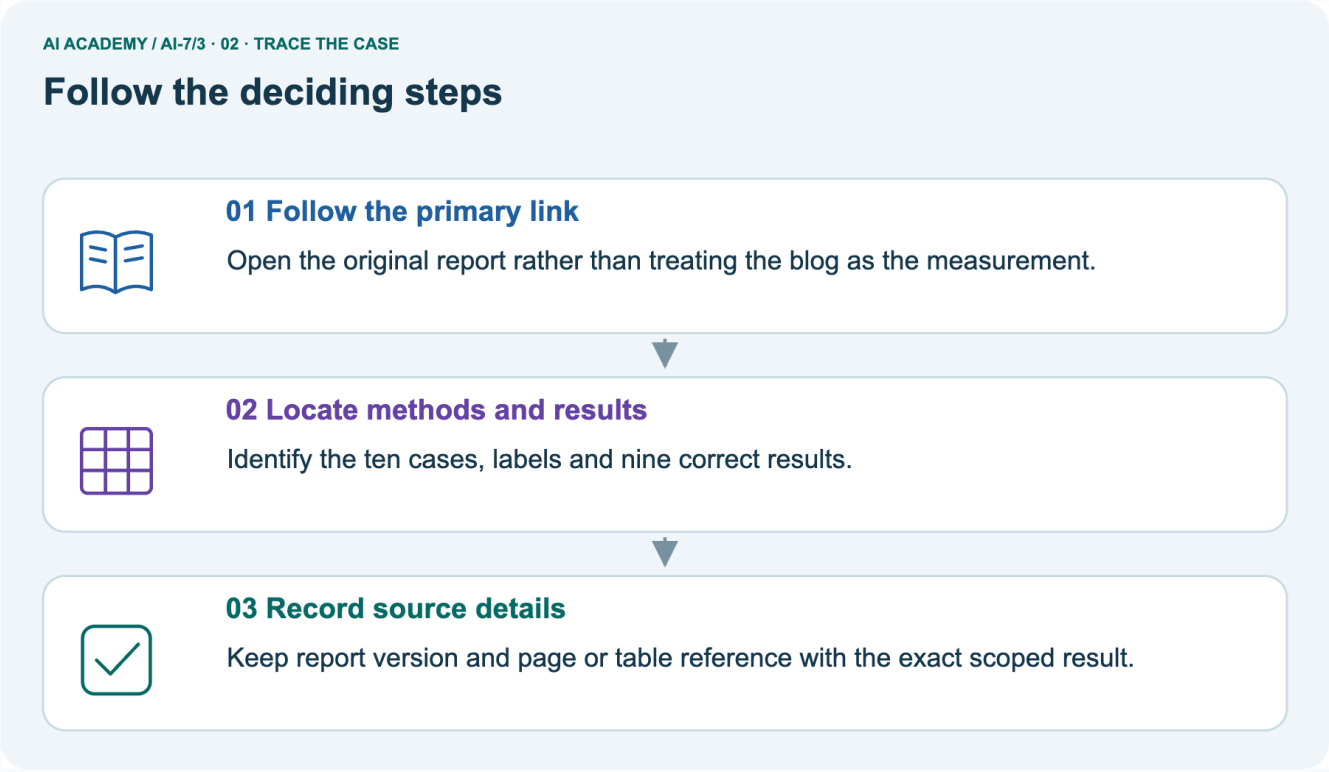
Find the source behind a summary. A blog describes a fictional benchmark improvement and links an original study report. Supplied facts: Blog says "90% reliable".

Original report records 9/10 correct on one fixed fictional task set and describes its labels.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the primary link: Open the original report rather than treating the blog as the measurement. → Locate methods and results: Identify the ten cases, labels and nine correct results. → Record source details: Keep report version and page or table reference with the exact scoped result.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Finding primary research — evidence trace

AI ACADEMY / AI-7/3 · 03 · INSPECT THE LESSON MODEL

Finding primary research — evidence trace

Input and scope	Check or calculation	Conclusion and limit
A blog says experiment had100 cases; original methods say40 and supplementary version says60 after revision.	Record title, authors, version, date, identifier and access location. Distinguish paper, code, data and commentary. Verify that a quoted claim belongs to the cited version.	Track versions and use the correct original record for the claim. The blog is secondary; the two original versions must not be merged silently.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the labelled picture with the matching explanation. Open the vector diagram

Finding primary research — **evidence** trace. A blog says experiment had100 cases; original methods say40 and supplementary version says60 after revision. | Record title, authors, version, date, identifier and access location.

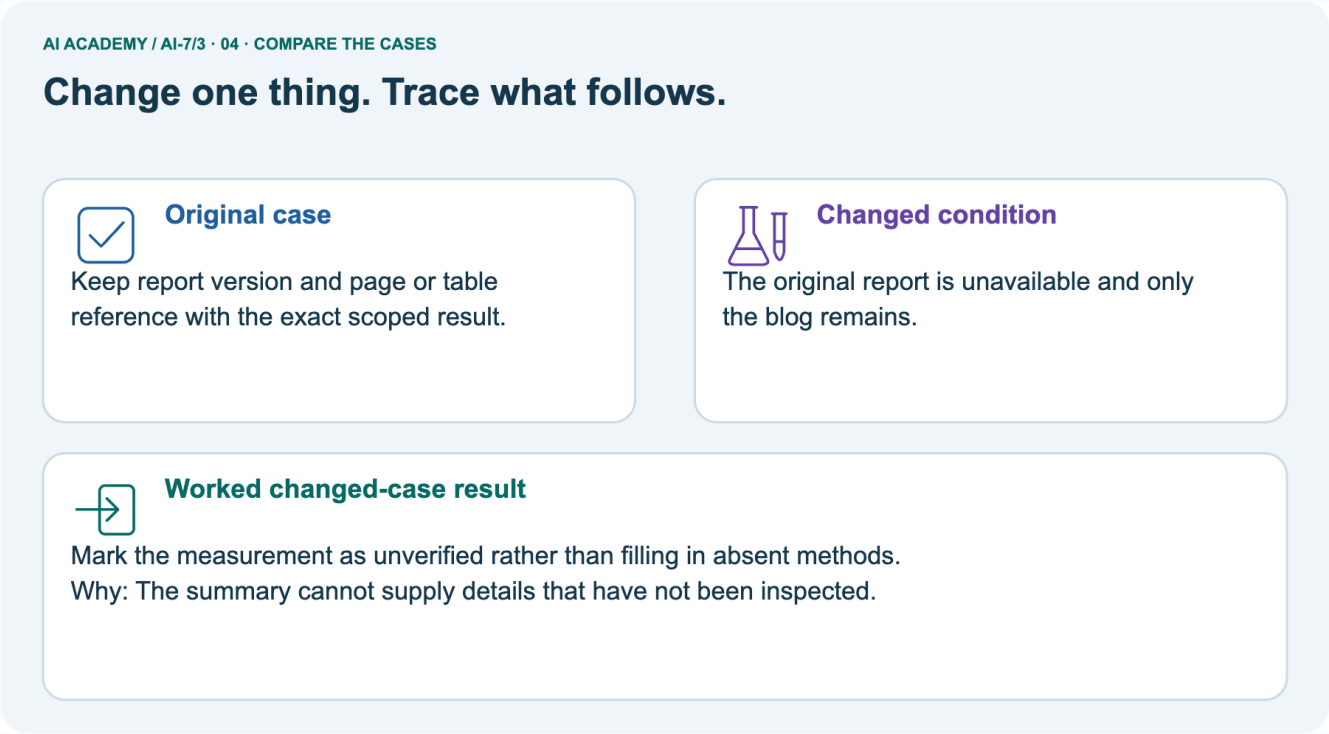
Distinguish paper, code, data and commentary. Verify that a quoted claim belongs to the cited version. | Track versions and use the correct original record for the claim.

The blog is secondary; the two original versions must not be merged silently.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

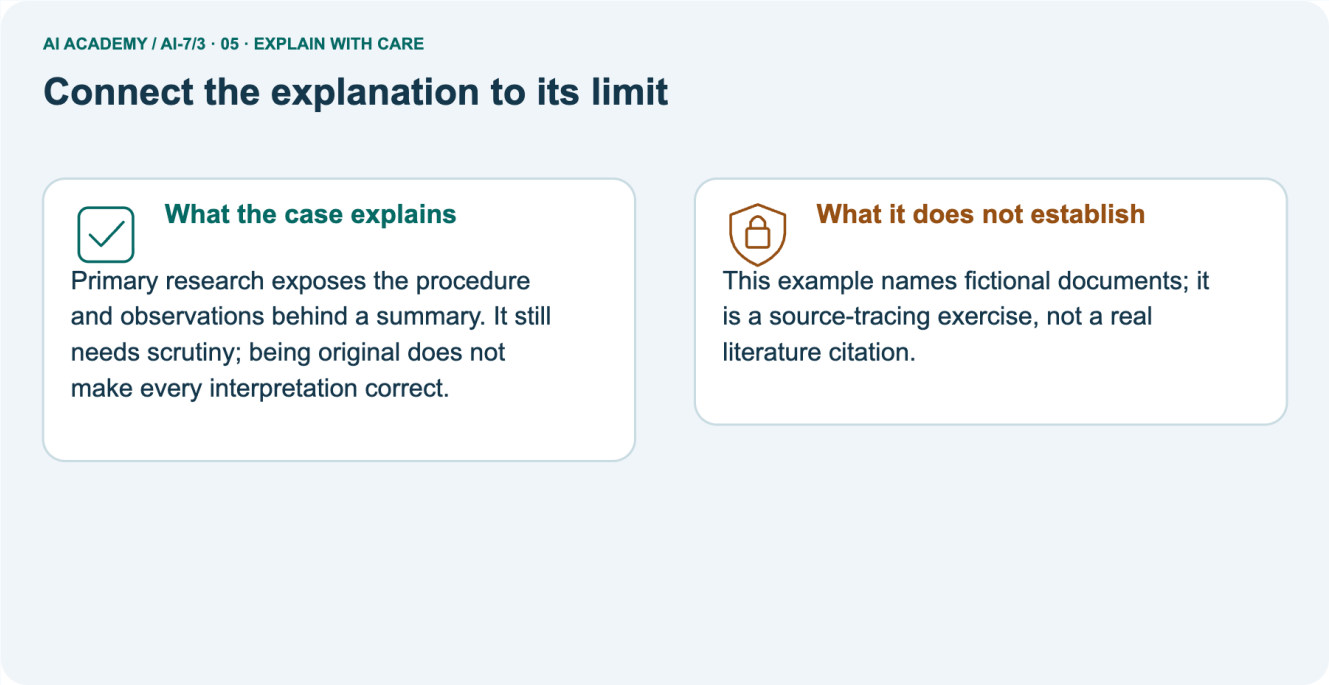
Original case: Keep report version and page or table reference with the exact scoped result.
Changed condition: The original report is unavailable and only the blog remains.

Expected result: Mark the measurement as unverified rather than filling in absent methods. Reason: The summary cannot supply details that have not been inspected.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

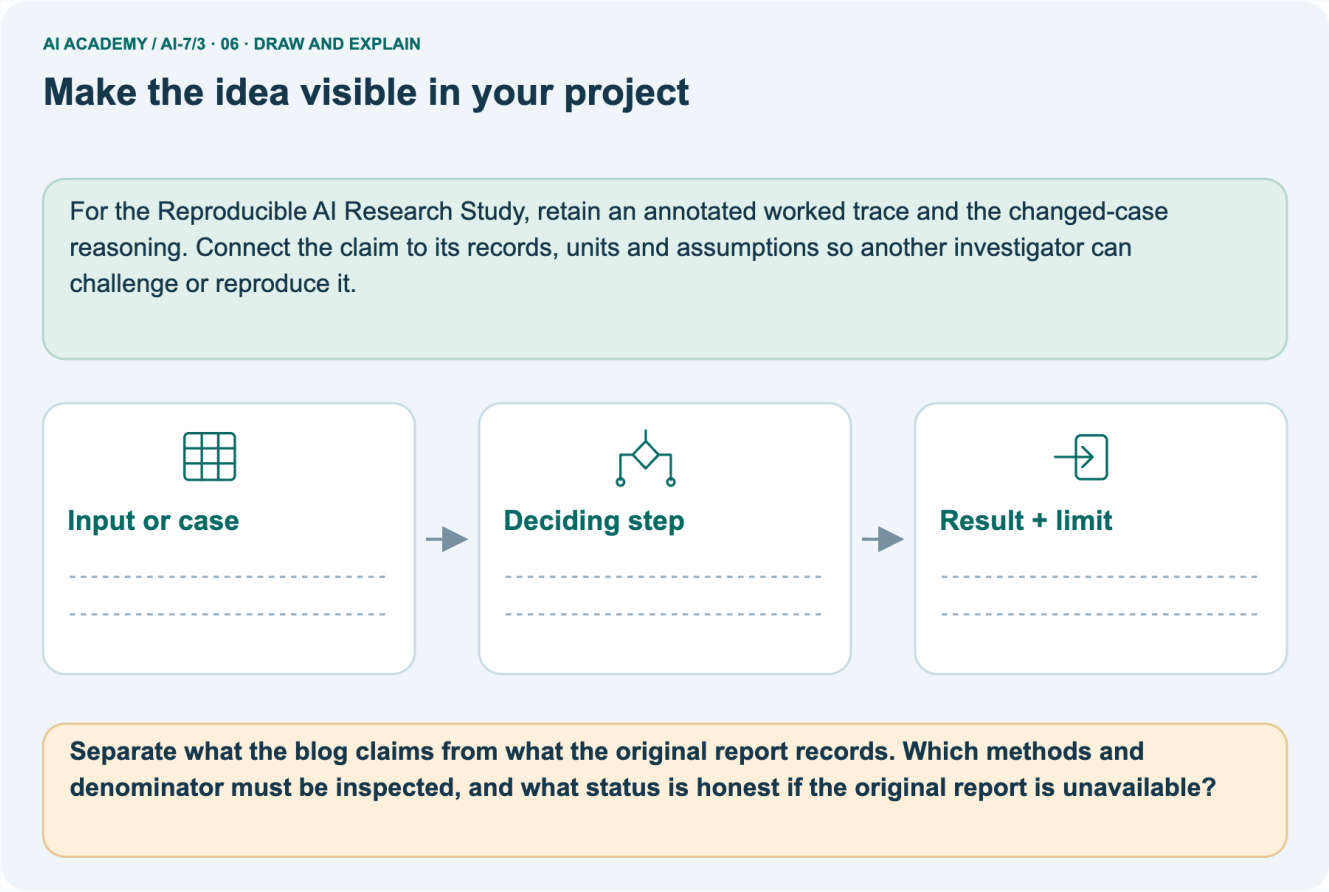
Primary research exposes the procedure and **observation**s behind a summary. It still needs scrutiny; being original does not make every interpretation correct.

Limit: This example names fictional documents; it is a source-tracing exercise, not a real literature citation.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Separate what the blog claims from what the original report records.

Which methods and denominator must be inspected, and what status is honest if the original report is unavailable?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.